

Writing a PhD application

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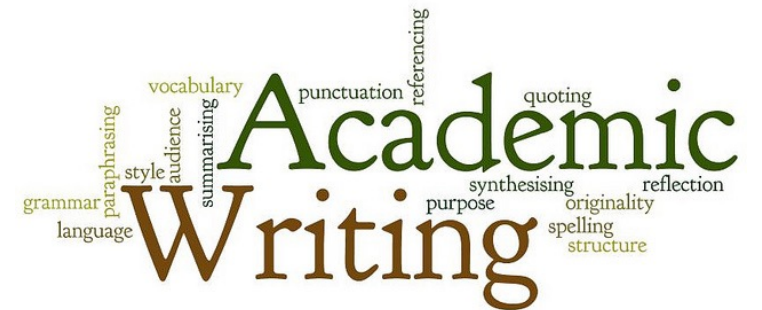
Developmental
Neuroscience and
Psychopathology, MRes



Anna Freud
building the mental
wellbeing of the
next generation

Session overview

1. Academic writing: considerations and tips
2. Academic writing: specifics of a PhD application
 - Personal statement
 - Research project
3. Practical
4. Assignment explanation



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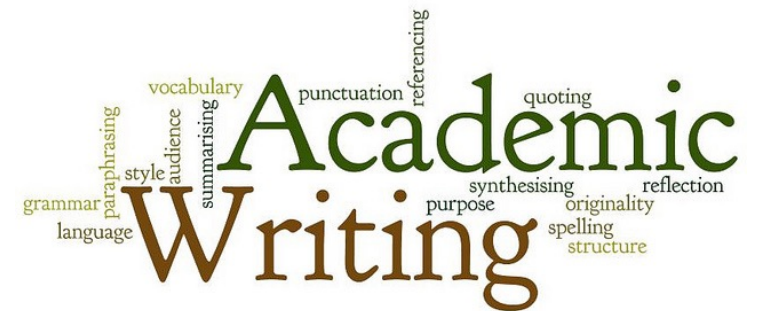
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Academic writing vs literature

- Academic writing is not literature
 - ❖ In literature, *how* you write is as important as *what* you write
 - ❖ In academic writing, the content comes first – writing style should be transparent
- Avoid:
 - ❖ Literary techniques (metaphors, adjectives, etc)
 - ❖ Fancy words
 - ❖ Technical jargon when not necessary
 - When technical concepts are necessary – explain them

Main function of academic writing

- Have a clearly defined message
- Deliver this message to your audience without any ambiguity
- Short sentences help you to do this
 - ❖ 10 word sentences are easy to comprehend; 20 words is ok; >30 words is bad!
 - ❖ When you have to read a sentence twice to understand it, it's too long
 - ❖ Check the word count of sentences and cut if too long
 - ❖ But vary sentence lengths to avoid boring the reader!

Consider your readers

- Define your readership and adapt your writing
 - ❖ E.g., no need to explain a t-test to experimental psychologists, but explain propensity score matching or multiple imputation
 - ❖ Explain the importance of your topic differently in a general vs speciality journal
- Readers will be varied in terms of their research topic and technical/statistical knowledge
 - ❖ Don't write for a niche
 - ❖ When in doubt, assume your reader needs explanations

Theory of mind for your readers

- Think of yourself as a reader:
 - ❖ How many articles do you find too long, unclear, and poorly written?
 - ❖ What makes you cringe – long sentences, acronyms, jargon, typos?
- Then imagine a real reader:
 - ❖ A psychologist you know working on a similar research topic
 - ❖ A person who may recruit you for a PhD / job (e.g., head of a panel)
- Consider what readers may find confusing or challenging about your writing, and adapt it to make it easy for them
 - ❖ A careful writer makes a happy reader!

Key concepts

- Explanation
 - ❖ Don't just make a point, *explain* it
 - ❖ Provide examples
- Structure: the hardest part!
 - ❖ Not just a list of sentences: points should logically follow from one another
 - ❖ The first sentence of each paragraph should provide the main point or argument, with additional detail provided within the body of the paragraph
 - ❖ There should be structure *within each paragraph* and *throughout the article*
 - ❖ At the beginning of the article/section, signpost to the reader what you will cover. E.g., "we will first discuss X, then Y"
- Awkwardness
 - ❖ Ungrammatical, unclear, difficult to follow
 - ❖ Use your own words - don't paraphrase or quote chunks from other authors

Writing rules to remember (S.P.C.C.C)

Specificity

Precision

Clarity

Consistency

Conciseness

Specificity

- Define a specific focus/purpose for your article / research proposal
- When writing or revising, eliminate all points that are not directly relevant to your main message
- You need to stay focused on the key message to keep readers' attention and make your case

Precision

- Scientific writing should be precise
 - ❖ Cut adjectives and imprecise terminology, e.g., “much more important”, “quite low
 - ❖ Stay neutral
- Quantify your claim (e.g., “large effect size of $r=0.7$ ” instead of “very important impact”)
- Clarify explicitly any judgement you make
 - ❖ E.g., “much more important than X because of X and X reasons”

Clarity

- Writing clearly is much more difficult than writing fancy sentences
 - ❖ Take the time you need! (It may be much more than you think)
- Use simple words
- Avoid “this”, “they”, “it”, or any word referring to something you mentioned before
 - ❖ E.g., “they were administered” -> “participants in condition A were administered”
 - ❖ “We controlled for covariates” -> “We controlled for family chaos and parenting”

Consistency

- There are often several words available for the same concept
 - ❖ E.g., covariates, controls, confounding variables
- Choose the word that fits your purpose best and use it consistently
- If not, readers will wonder whether there are subtle differences in the concepts you're using -> distracting
- Repetition is ok!
 - ❖ It's not literature

Conciseness

- Word counts force you to be concise
- Think carefully about the utility of every sentence and every word
- Factor in several painful hours to respect the word count limit
- Omit unnecessary words, e.g.:
 - ❖ “In future studies, researchers may want to address” -> “Future studies may address”
 - ❖ “The purpose of the current study was...” -> “We aimed to”
 - ❖ “To date, there have been minimal studies that examined” -> “To date, few studies examined”

Paragraphs

- Each paragraph should contain one idea or main point
- The first sentence of each paragraph should introduce the point, to be elaborated on within the paragraph
 - ❖ Known as the “topic sentence”
- Readers should be able to skim the entire text, reading only the paragraph topic sentences to understand the general meaning

Finding and planning your story means accomplishing three things. First, you must identify your central question and its answer (the 2-sentence minisummary from the last section). Write this down, even though no reader will ever see it; the writing will force you to be explicit. Second, you must decide which information, data, analyses, and interpretation belong in the paper, and which are better reserved for another manuscript or abandoned to a dusty row of old notebooks. Third, once you've identified the content, you must decide in what order it should be presented. Of these, the first point is finding a story to tell, and the second and third are planning the best way to tell it.

Because the path from concept through execution to data analysis and interpretation is rarely simple, finding your story uses hindsight as much as foresight. Writers who don't realize this often cling to reporting everything they did, in the order they did it, including experiments that turned out to be blind alleys and observations that seemed relevant at the proposal stage but became immaterial to the conclusions eventually drawn. Experienced writers, in contrast, take full advantage of hindsight and devote a lot of labour to deciding on the story they want the reader to hear (whether or not it's the story they had in mind when they began the work), and what information the reader needs to understand that story.

You should not hope to find your story by sitting down at your keyboard and beginning to type the first line of your Introduction. Instead, you should begin by brainstorming possible content and then selecting and organizing the content that defines your story and tells it effectively. There are many techniques for finding and planning your story; what follows is a toolbox from which you can select. Many writers apply more than one of these techniques to each writing project.

Transitions

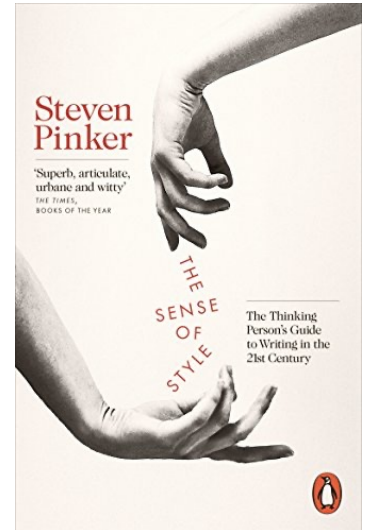
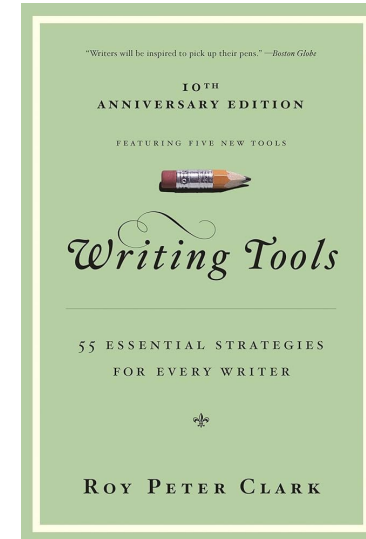
- Use transition statements, e.g.:
 - ❖ “Turning now to the evidence on...”
 - ❖ “Regarding the finding that...”
- Use bridge words (same word at the end of a paragraph and at the beginning of the next one) to link:
 - ❖ 1st paragraph: “However, it is not clear if social support has a casual protective effect, as most research has not accounted for **genetic confounding**.”
 - ❖ 2nd paragraph: “**Genetic confounding** might arise if genetic influences on mental health problems affect a child’s exposure to social support...”

Sentences

- Vary sentence opener (“however”, “on the other hand”, “for instance” etc...)
- Use the active voice:
 - ❖ “It was found that” -> “We found that”
- Use positive form:
 - ❖ “It is not clear” -> “it is unclear” (one word saved!)

Training in writing well

- Learning to write well takes time, practice, and effort
- Resources are available with expert writing tips
- Tips to improve your writing:
 - Start a document with corrections & useful expressions
 - Notice good and bad writing when you read
 - When writing, consult other articles that are written well and follow a similar structure
 - Join writing clubs if available



Revising your writing

- No writer can write clearly and concisely right from the start
- Revising is key: it's normal to have many drafts!
- Reading your writing out loud can help you spot any awkward sentences or grammatical errors
- It can be hard if you send it to feedback and receive it covered in tracked changes – but this is helpful overall!
 - ❖ Ask your supervisor / reader to explain the changes so you understand why and how to improve for the future

Resources

- Many useful expressions and sentences on academic phrasebank:
- <http://www.phrasebank.manchester.ac.uk/>
- Also explore the “useful links” on the site.

Newcastle University Academic Writing Guide lists a lot of resources and references on writing:

<http://www.ncl.ac.uk/students/wdc/learning/academic/>

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Expectations for funded PhD applications

- Education
 - ❖ First-class or 2.1 degree – due to high competition, most panels will expect a first-class degree, or a 2.1 with Masters at merit or distinction level
- Some previous research experience
 - ❖ Research assistant role
 - ❖ Presentation of posters at conferences or co-authorship in articles is great
- Specifics for the project
 - ❖ Specific statistical training, broad understanding of the topic, demonstrate specific interest

Types of funded PhD programme

- 1+3 programme (MSc + PhD):
 - ❖ Requires a personal statement
 - ❖ Not necessarily a research proposal (projects provided by supervisors)
- Already funded project (e.g., as part of a PI's grant):
 - ❖ Requires a personal statement
 - ❖ Not necessarily a research proposal (as PI already has the project planned)
- Apply for funding:
 - ❖ Requires a personal statement
 - ❖ Requires a research proposal (written in collaboration with potential supervisor)

Identify requirements

Gather information on:

- University and department
- Potential supervisors
- Specific research project

If applying to an already funded PhD project, study the job description:

- What kind of academic profile do they want?
- What previous research experience: data collection, neuroimaging, analysis?
- Specific technical skills: databases, statistical software?
- Any personal skills (e.g. working in team, contact with people?)

Identify requirements (2)

- Contact people to understand more about the projects/environment:
 - ❖ Former students
 - ❖ Potential supervisors
 - ❖ Person in charge of programs
 - if you decide to contact them, be polite and have specific questions.
- E.g., at the Institute of Cognitive Neuroscience, UCL, you need to contact the research group leader to plan a potential project
 - ❖ Brief outline of your interests/research plans
 - ❖ CV
- Once you have identified requirements for application, address all of them.

Personal statement

- Don't repeat your CV!
- Link your qualifications and interests to the position requirements in a direct and personal way
- 1. Explain what you can bring to the research project
- 2. Explain specific elements of the university/department/research team that appeal to you.

Personal statement: elements

- Carefully read and strictly follow all requirements in the application forms
- Briefly describe your academic background, highlighting:
 - Your strengths (if first class, any other award etc.)
 - Any research experience:
 - General
 - Anything specific that corresponds to the project
- Personal characteristics
 - That make you suitable for research...
 - ...and this research project in particular

Personal statement: elements (2)

- Briefly mention your research interests (more important when you are not applying to a specific research project)
- Think strategically: how broad your stated research interests are will depend on where you apply:
 - E.g. for a specific funded project: focus on the research project
 - E.g. in some programmes, you have to choose 3 potential projects from a list: they like you to be quite flexible about what you are ready to do
 - Remember that your actual research may be quite different from your project

BE SPECIFIC: most of what you did is interesting, not all is relevant!

Personal statement: tone

- Don't be modest: all programs and universities are explicitly looking for “excellent”, “outstanding” etc. candidates.
- But make sure your claims are substantiated!
- Examples are key:
 - ❖ I am productive: I have presented a poster at a conference...
 - ❖ I'm patient: I have cleaned data bases...
 - ❖ I am creative: My undergrad research project was awarded...
 - ❖ My skills are relevant for the project: I have done data collection in primary schools

Research proposal

- Research project should be written in collaboration with your future PhD supervisor
 - ❖ Get as much feedback from them as you can!
- Read and follow any instruction on the content and the form of your research project:
 - ❖ Length
 - ❖ Structure
 - ❖ Number of references...

Research proposal - originality

- Your research should be original:
 - ❖ Area that has rarely been explored
 - ❖ Address important gaps in the literature
 - ❖ Under-studied population
 - ❖ Use innovative methods and/or data analysis techniques
- Address what makes your project original explicitly
- Reviewers don't have time to work it out!

Research proposal - credibility

- Reviewers will assess the project's credibility and feasibility
 - ❖ Can you deliver on participants' recruitment?
 - ❖ Are measures and procedures described in sufficient details?
 - ❖ Do you or the team you'll be in have the required methodological skills?
 - ❖ If not, how will you acquire them? E.g. training
 - ❖ Is your project feasible within the budget and timeframe?

Research proposal - structure

- Background/literature review:
 - Why is your topic important (try to be specific)
 - What is known on the subject
 - Limitations of existing studies
- Rationale and objectives:
 - This research will address gaps in the literature (be explicit!)
 - Clear set of objectives
- Methods:
 - Participants/data collection
 - Measures
 - Data analysis

Final comments

- Personal statement and/or research projects are considered writing samples:
 - ❖ You will be assessed not only on the content but also on the form
 - ❖ Should be well written and structured
 - ❖ Free of all grammatical and spelling errors
- Everything has to be carefully crafted: a very good research project with a bad personal statement will not get you an interview
- There may be 100 applications for 2 positions
 - ❖ Be explicit about what makes your application original and excellent!

Interview

- Prepare a lot – anticipate some general questions
 - ❖ Why this project/programme/university?
 - ❖ Explain your previous research experience
 - ❖ Why would you be a good fit for this project / programme?
- Know the literature; read all recommended papers
- The main criterion is whether you can speak intelligently of your research project and research in general, meaning:
 - ❖ Knowledge
 - ❖ Critical thinking
 - ❖ Argumentative capacities (they might criticise your point of view to test how you react)
 - ❖ Have your own opinions but be nuanced
 - ❖ Be open-minded about which research project you would be willing to do

Feedback

- Science is about collaboration – get feedback on your proposal & personal statement:
 - ❖ From your fellow students
 - ❖ From your future supervisors
 - ❖ From friends, family (spelling mistakes, can a lay person understand at least broadly your topic?)
- If you get an interview for a project you applied for with a supervisor, ask them to help you arrange a mock interview
 - ❖ Or ask you some practice questions themselves in a mock!

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Practical: personal statement

- Check each of the following aspects and note what can be improved:
- Highlight strengths:
 - Relevant research experience and personal characteristics
 - Strategic presentation of research interests
 - Specific links between the profile and the institution/department/project
 - All claims are justified and illustrated
 - Style: clear, convincing, without grammatical and spelling errors

Practical: research project

- Check each aspect:
- Elements of structure
- Originality and credibility
- Writing rules followed?
 - Specificity
 - Precision
 - Clarity
 - Consistency
 - Conciseness

Research proposal (1)

- Structure:
 - Background: clear overview of the literature and identification of the current limitation
 - Clear overarching topic with specific studies
 - Very thorough descriptions of measures but not of analyses. Intervention not clear.
 - Not much on analysis. No power analysis, problematic for candidate gene analysis.
- Originality and credibility:
 - Explicit that this is a large study on an under-investigated topic
 - Feasibility: collaboration with an experienced association
- Style:
 - Short sentences, very clearly laid out.
 - A few typos.

Research project (2)

- Structure:
 - Background: overview of the literature and identification of the current limitations
 - Not so clear how the theoretical models relate to the actual study
- Originality and credibility:
 - Explicitly states what is original
 - Not many elements of credibility: can the student deliver, is it the right place?
- Style:
 - Some sentences could be more concise
 - A few typos, e.g. “in addiction”

Improving your application

- Would you select this applicant? Would you select yourself?
- What can you do in the next 6 months to improve your application?
 - Additional research experience
 - Presenting a poster to a conference or trying to participate in a paper?
 - Statistical training?
- Keep working at your application:
 - Do not copy and paste, really adapt it for all your applications
 - Apply for your own funding if possible: highly valued
 - www.jobs.ac.uk

Assignment: Research proposal

- Choose a research topic that corresponds to your research experience/interests:
 - Longitudinal perspective
 - Children and/or adolescents
 - Quantitative
 - Innovative
 - 2/3 studies under an overarching theme
 - Prepare a research proposal (80% of the overall mark, max total words 2,200):
 - A. Background' section including the following sub-headings: 'Literature Review', 'Rationale' and 'Main Objectives' (total maximum 1,000 words)
 - B. 'Proposed PhD Research' section comprising the following sub-headings for each study: 'Objective', 'Participants', 'Methods', 'Predictions'. (1,200 words)
 - References: No more than 15 references in total in APA style. Note references are not included in the word count. In the text, references may be numbered.
- IMPORTANT: points for writing and presentation

Assignment: Personal statement

- Personal statement (20% of the overall mark):
 - ❖ No more than 600 words
 - ❖ To make it realistic: select an institution (e.g. Yale) and identify a potential supervisor
 - ❖ No imposed structure, but you should address the following points:
 - ❖ Reasons for applying
 - ❖ Relevant experience for the project, strengths and personal qualities
 - ❖ Conclusion

LOOK AT DETAILED INSTRUCTIONS: PART OF THE EXERCISE

References

- *Silvia P. J. (2007) *How to write a lot. A practical guide to productive academic writing*. American Psychological Association. Washington, DC.
- **Chen** et al. (2008) Reflections on becoming a successful researcher, *Educ Psychol Rev*, 20:65-70.